

ELASTIC CACHE WITH AWS

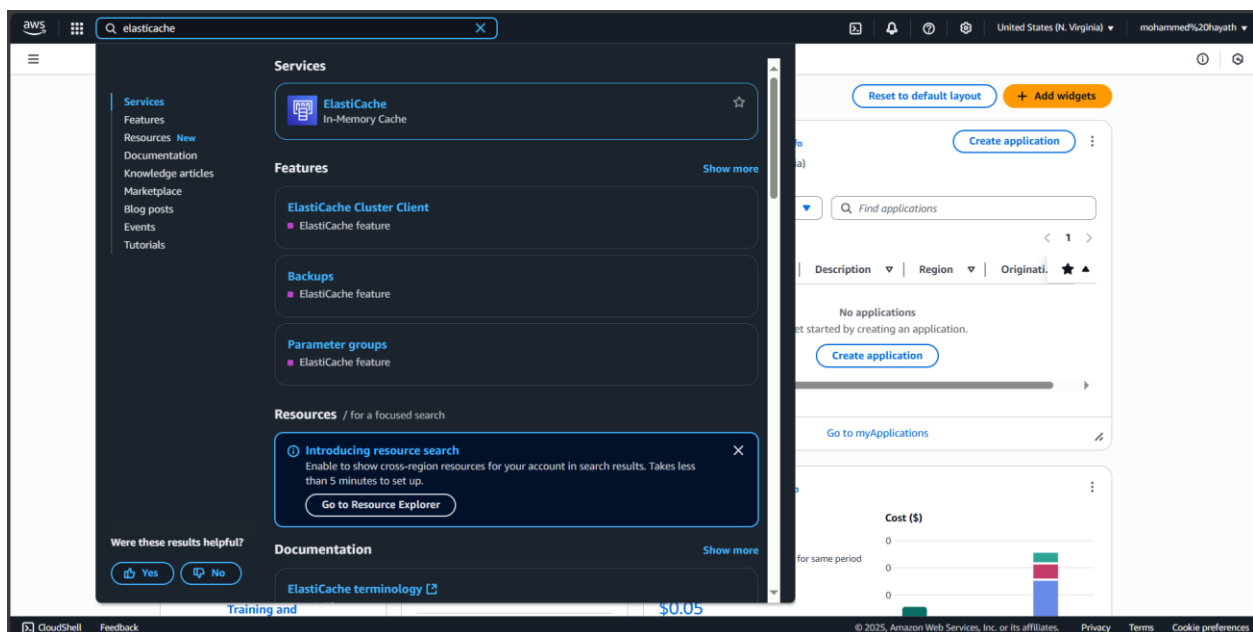
What is Amazon ElastiCache?

Welcome to the Amazon ElastiCache User Guide. Amazon ElastiCache is a web service that makes it easy to set up, manage, and scale a distributed in-memory data store or cache environment in the cloud. It provides a high-performance, scalable, and cost-effective caching solution. At the same time, it helps remove the complexity associated with deploying and managing a distributed cache environment.

Note

Amazon ElastiCache works with the Valkey, Memcached, and Redis OSS engines. If you're unsure which engine you want to use, see [Comparing Valkey, Memcached, and Redis OSS self-designed caches](#) in this guide.

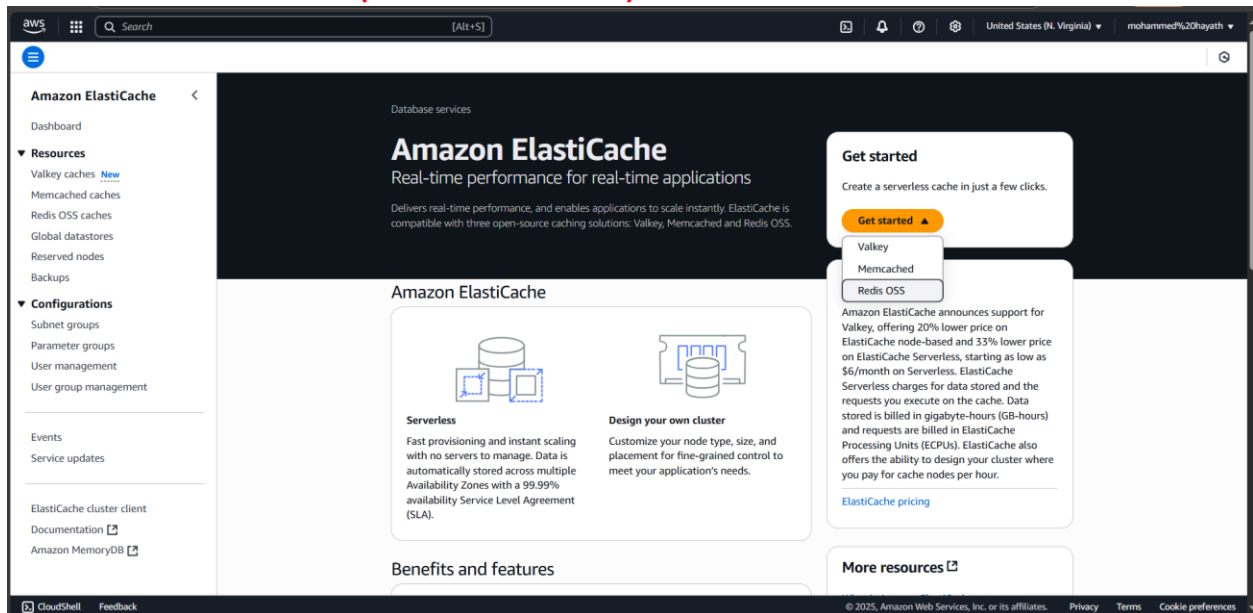
Go to aws Elastic cache Service



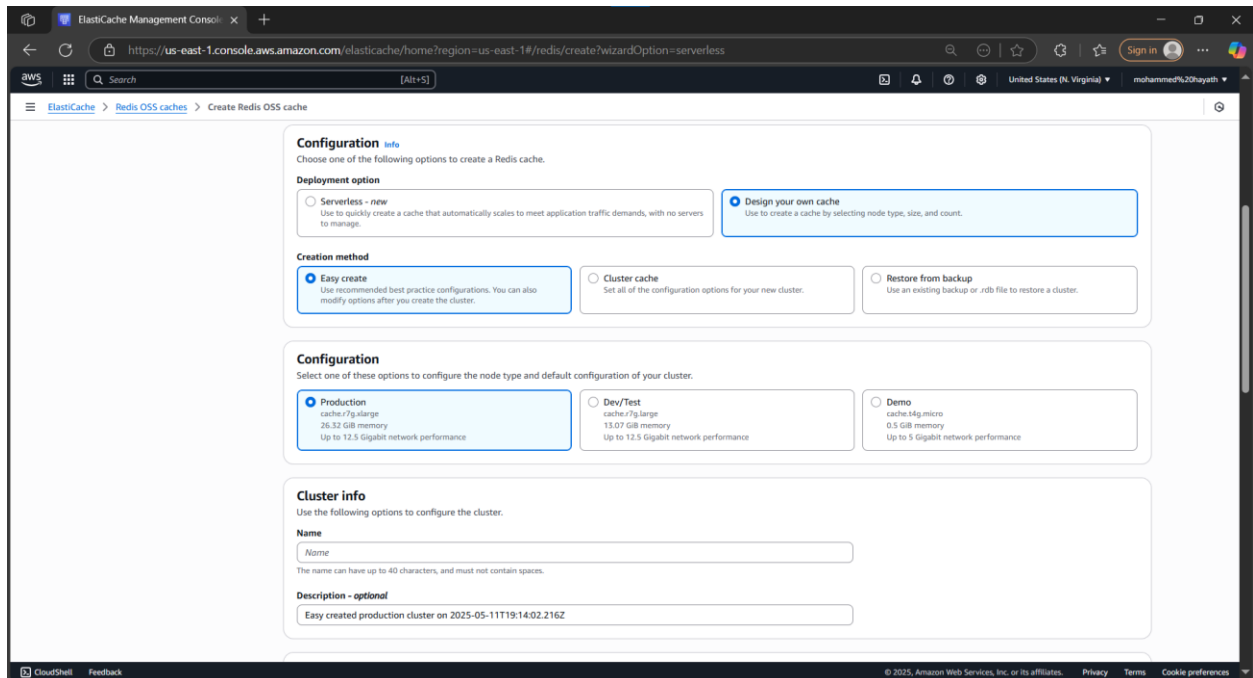
Create a redis cluster

ELASTIC CACHE WITH AWS

Choose redis OSS (best in cache)

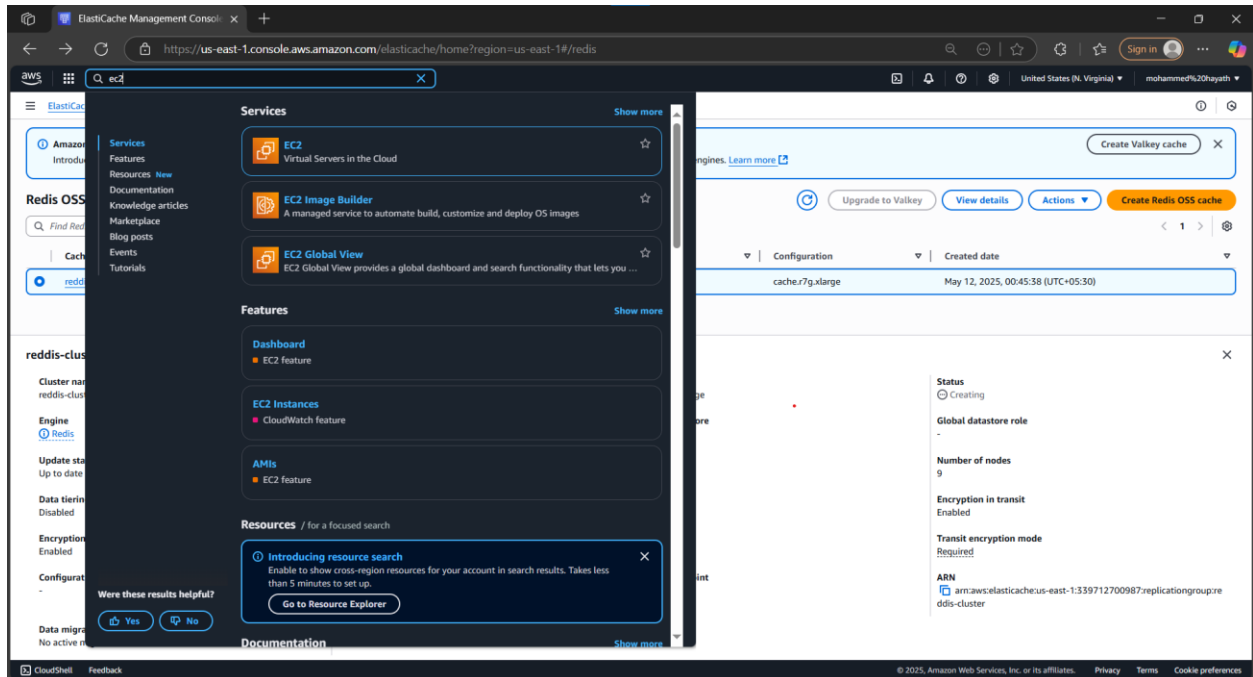


Select serverless or Design your own (as your wish)

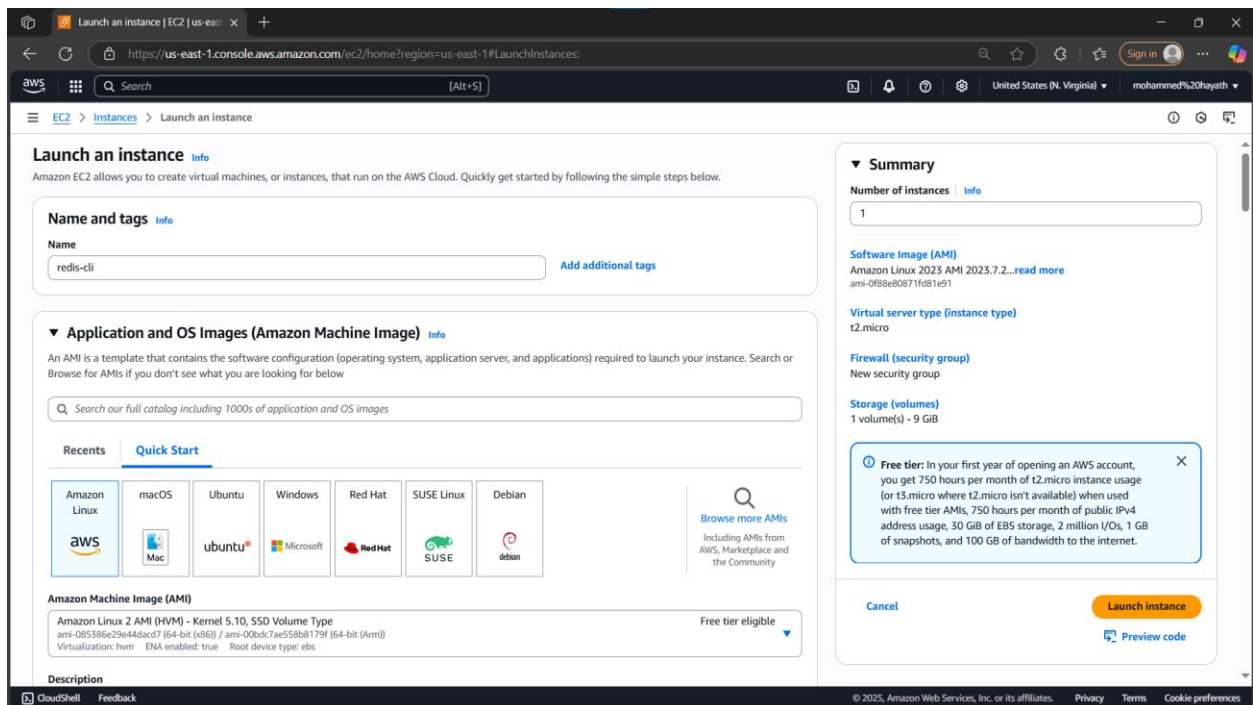


ELASTIC CACHE WITH AWS

create EC2 to connect to your redis cache



Select aws linux OS for better connectivity



ELASTIC CACHE WITH AWS

Configure network settings and enable inbound with port range 6379
And create new security group

Launch an instance | EC2 | us-east-1

<https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances>

Network settings Info

VPC - required Info
vpc-0a2823024191442d5 (default) 172.31.0.0/16

Subnet Info
subnet-04a3325d3141b7754 VPC: vpc-0a2823024191442d5 Owner: 339712700987 Availability Zone: us-east-1b Zone type: Availability Zone IP addresses available: 4089 CIDR: 172.31.80.0/20 [Create new subnet](#)

Auto-assign public IP Info
Enable [Additional charges apply when outside of free tier allowance](#)

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☒ Create security group ☐ Select existing security group

Security group name - required
launch-wizard-3
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _./:/@.[!+=&(),]*

Description - required Info
launch-wizard-3 created 2025-05-11T19:17:47.378Z

Inbound Security Group Rules
▼ Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type	Protocol	Port range
ssh	TCP	22

[Remove](#)

Summary
Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-0f88e80871f081e91

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 9 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

Instances | EC2 | us-east-1

<https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running>

Instances (1) Info
Find Instance by attribute or tag (case-sensitive) All states

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
redis-cli	i-0df64e068642df925	Running	t2.micro	Initializing	View alarms +	us-east-1b	ec2-13-218-131-235.co...	13.218.

Select an instance

ELASTIC CACHE WITH AWS

Connect your EC2 from your terminal

```
[ec2-user@ip-172-31-91-237:~  
C:\Users\Admin\Desktop\key-pair>ssh -i "C:\Users\Admin\Desktop\key-pair\redis-keypair.pem" ec2-user@13.218.131.235  
Warning: Identity file C:\Users\Admin\Desktop\key-pair\redis-keypair.pem not accessible: No such file or directory.  
ec2-user@13.218.131.235: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
  
C:\Users\Admin\Desktop\key-pair>cd ..  
  
C:\Users\Admin\Desktop>cd ..  
  
C:\Users\Admin>cd downloads  
  
C:\Users\Admin\Downloads>ssh -i red.pem ubuntu@35.175.148.110  
The authenticity of host '35.175.148.110 (35.175.148.110)' can't be established.  
ED25519 key fingerprint is SHA256:2iPaKhWRSC48IOJSRS+xYlXtoOPPKj9b0dRycBDwXwY.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '35.175.148.110' (ED25519) to the list of known hosts.  
ubuntu@35.175.148.110: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
  
C:\Users\Admin\Downloads>ssh -i red.pem ec2-user@35.175.148.110  
  
#_ _  
~ \ ##### Amazon Linux 2023  
~~~ \#####\  
~~~~ \###|  
~~~~ \#/ _____ https://aws.amazon.com/linux/amazon-linux-2023  
~~~~ V~' '->  
~~~~ ~~~~~  
~~~~ / / / / /  
~~~~ /m/'
```

```
[ec2-user@ip-172-31-91-237 ~]$
```

Install all these packages in your Ec2 instace to build redis cache

1. Installs the Extra Packages for Enterprise Linux (EPEL)

- Installs the Extra Packages for Enterprise Linux (EPEL) repository on an Amazon Linux system. EPEL provides additional software packages that are not included in the default Amazon Linux repositories.

```
```bash
sudo amazon-linux-extras install epel -y
```
```

2. install required packgaes and dependencies

- Installs various development libraries and tools that are necessary for building and running Redis. These include the GNU Compiler Collection (GCC), Jemalloc (a memory allocator), OpenSSL development libraries, and Tcl (Tool Command Language) along with its development libraries.

```
```bash
sudo yum install gcc jemalloc-devel openssl-devel tcl tcl-devel -y
```
```

ELASTIC CACHE WITH AWS

3. Download Source Code and unzip

- Downloads the Redis source code archive (`redis-stable.tar.gz`) from the official Redis website.

```
```bash
sudo wget http://download.redis.io/redis-stable.tar.gz
```
```

```
```bash
sudo tar xvzf redis-stable.tar.gz
```
```

```
```bash
cd redis-stable
```
```

4. build redis

- Builds Redis from the source code. The `BUILD\_TLS=yes` option indicates that Redis should be built with support for Transport Layer Security (TLS), which is used for secure communication.

```
```bash
sudo make BUILD_TLS=yes
```
```

5. Grant Executable Permissions

- Grants execute (`+x`) permission to the Redis command-line interface (`redis-cli`) binary. This allows the binary to be executed as a program. Copies the Redis CLI binary (`redis-cli`) to the `/usr/bin/` directory. Placing it in this directory allows you to run `redis-cli` from any location in the terminal without specifying the full path.

```
```bash
chmod a+x /src/redis-cli
```
```

```
```bash
```

# ELASTIC CACHE WITH AWS

```
cp /src/redis-cli /usr/bin/
```
```

6. Connect to Redis Cluster

- Invokes the Redis command-line interface, connecting to a Redis server with the following options:

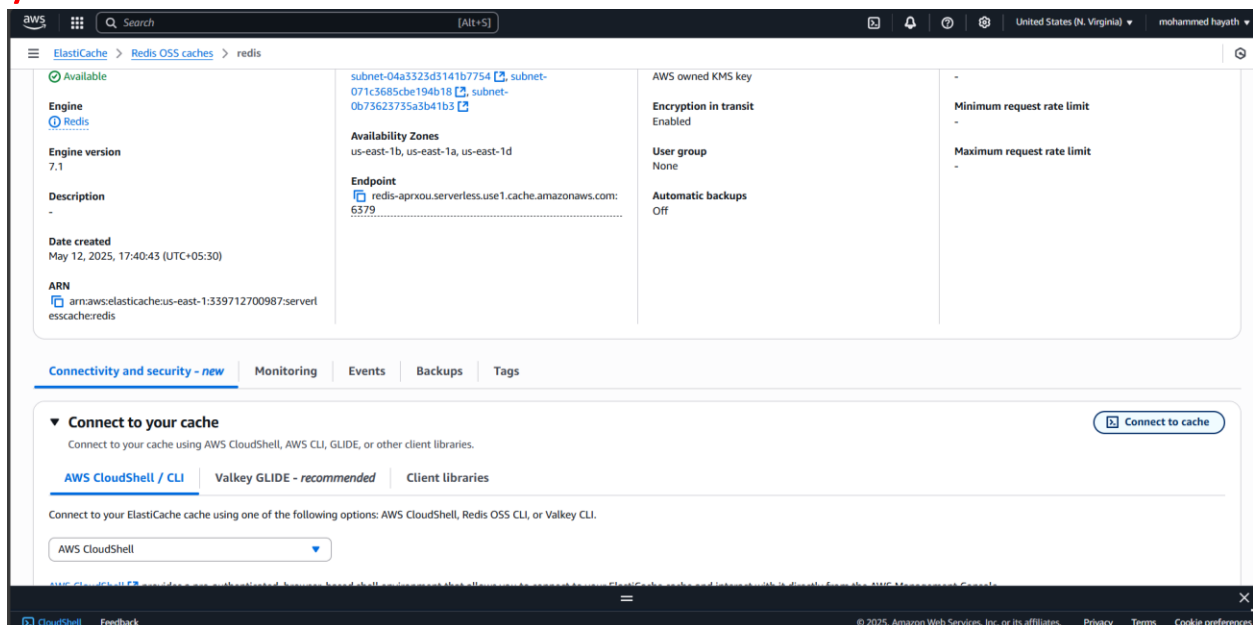
- `-h`: Specifies the hostname or IP address of the Redis server (``your-cluster-info.cache.amazonaws.com``).
- `-tls`: Enables TLS for secure communication.
- `-a`: Provides the authentication password (``YOURPASSWORD``).
- `-p`: Specifies the port number (``6379``) on which the Redis server is running.

```
```bash
```

```
redis6-cli --tls -h redis-aprxou.serverless.use1.cache.amazonaws.com -p 6379
```
```

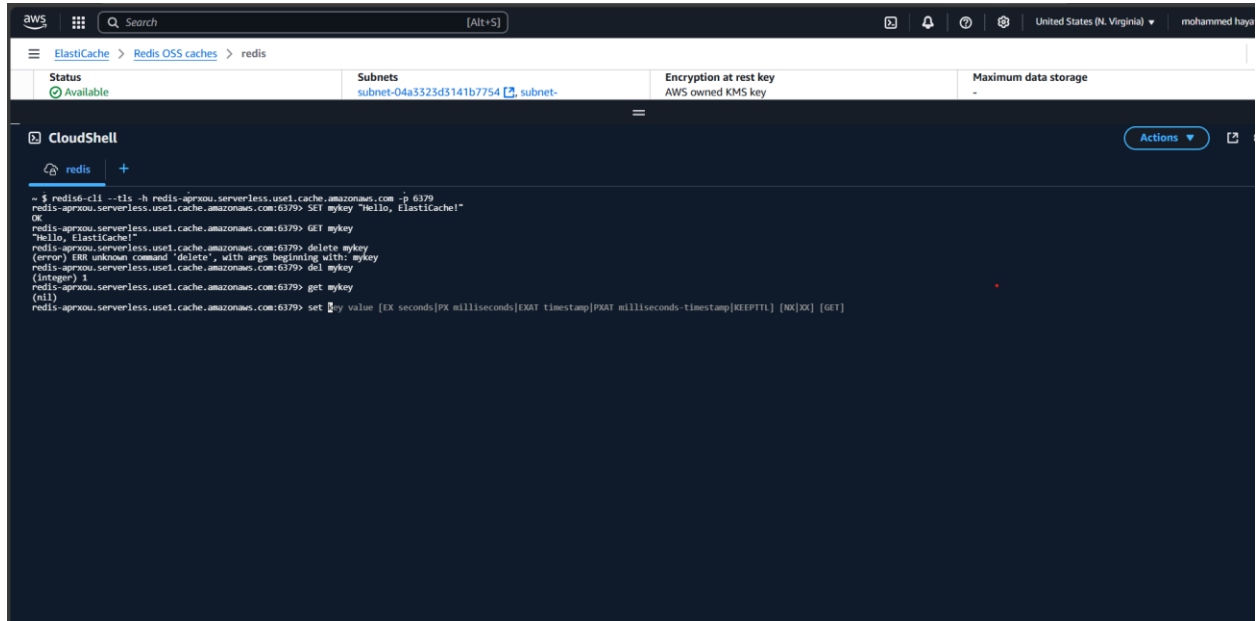
OR

you can be able to connect from aws console also

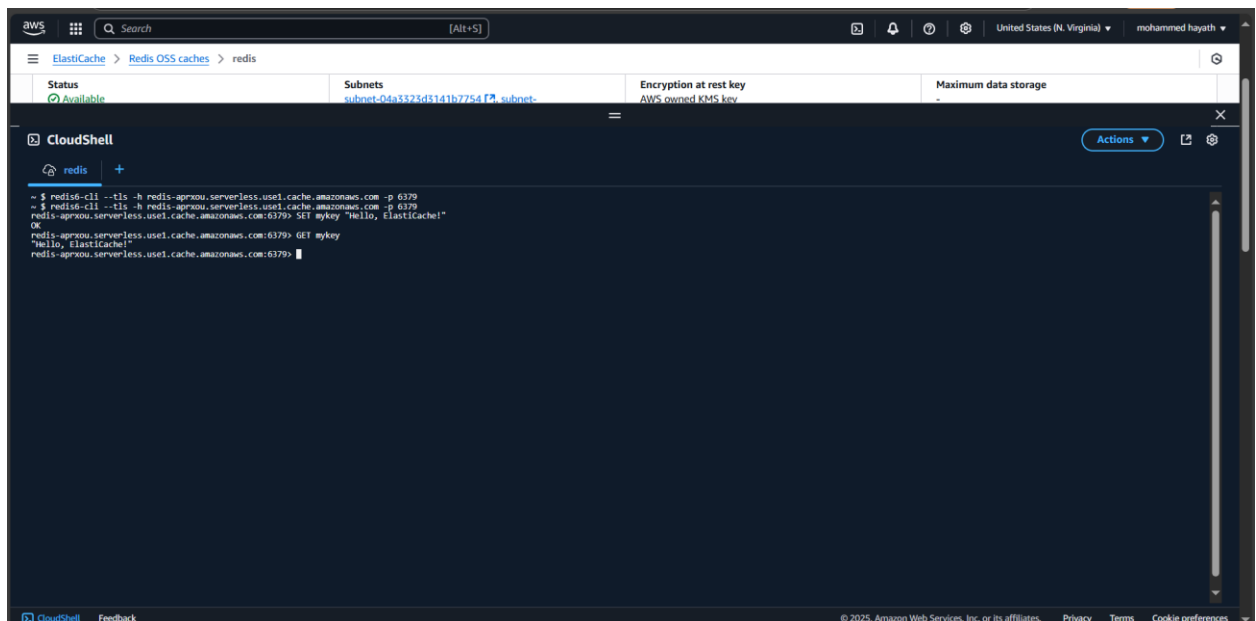


ELASTIC CACHE WITH AWS

Aws console cli



```
~$ redis-cli -h redis-aprxou.serverless.usel.cache.amazonaws.com -p 6379
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> SET mykey "Hello, ElastiCache!"
OK
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> GET mykey
"Hello, ElastiCache!"
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> delete mykey
(error) ERR unknown command 'delete', with args beginning with: mykey
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> del mykey
(integer) 1
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> get mykey
(nil)
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> set mykey value [EX seconds|PX milliseconds|EXAT timestamp|PXAT milliseconds timestamp|KEEPTTL] [NX|XX] [GET]
```



```
~$ redis-cli -h redis-aprxou.serverless.usel.cache.amazonaws.com -p 6379
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> SET mykey "Hello, ElastiCache!"
OK
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> GET mykey
"Hello, ElastiCache!"
redis-aprxou.serverless.usel.cache.amazonaws.com:6379> del mykey
(integer) 1
```